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Oefeningenreeks 4A nr 18.4

$$\begin{aligned} & \int x^3 \cos x dx \\ & \left\{ \begin{array}{l} u = x^3 \\ dv = \cos x dx \end{array} \right. \Rightarrow \left\{ \begin{array}{l} du = 3x^2 dx \\ v = \sin x \end{array} \right. \\ & = x^3 \sin x - 3 \int x^2 \sin x dx \\ & \left\{ \begin{array}{l} u = x^2 \\ dv = \sin x dx \end{array} \right. \Rightarrow \left\{ \begin{array}{l} du = 2x dx \\ v = -\cos x \end{array} \right. \\ & = x^3 \sin x + 3x^2 \cos x + 6 \int -\cos x dx \\ & \left\{ \begin{array}{l} u = x \\ dv = -\cos x dx \end{array} \right. \Rightarrow \left\{ \begin{array}{l} du = dx \\ v = -\sin x \end{array} \right. \\ & = x^3 \sin x + 3x^2 \cos x - 6x \sin x - 6 \int -\sin x dx \\ & = x^3 \sin x + 3x^2 \cos x - 6x \sin x - 6 \cos x + C \end{aligned}$$

References